

# Single-Pass Scaling for Motion Forecasting with Bayesian Last Layers

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**Abstract.** Uncertainty in human motion forecasting has two qualitatively different components: (i) aleatoric ambiguity in future dynamics and (ii) epistemic uncertainty caused by limited training support (rare motions, unseen subjects, distribution shift). Many Bayesian deep learning approaches estimate epistemic uncertainty via Monte Carlo sampling or ensembles [15, 26], which are often expensive and do not provide distribution-free finite-sample coverage guarantees without an additional calibration layer [14, 48]. We propose a lightweight *conjugate Bayesian last-layer* construction that yields a closed-form epistemic scaling factor  $\kappa(x)$  (or timewise  $\kappa_t(x)$ ) on top of a strong multilayer perceptron (MLP) feature extractor, and we combine it with a structured Gaussian uncertainty head that models spatio-temporal correlations. Finally, we apply split conformal prediction to achieve calibrated 95% prediction tubes with finite-sample marginal validity under exchangeability. Beyond global tube calibration, we also introduce  $\kappa$ -conditioned (Mondrian) conformal scaling and a trajectory-level structured conformal ellipsoid. Across nine datasets and deterministic, multimodal, and calibration baselines, the hybrid retains competitive point accuracy with strong likelihood and calibrated uncertainty. Across nine datasets, the recommended default is *ConjGraph-CP*, combining a MatrixNormal-GraphJ structured Gaussian head with time-coupled  $\kappa_t$  shrinkage and achieving the best overall MPJPE+NLL rank while producing calibrated 95% tubes with strong efficiency. Moreover,  $\kappa$  provides a lightweight epistemic risk score that stratifies error and can trigger conservative fallbacks.

**Keywords:** Human motion forecasting · uncertainty quantification · Bayesian last layer · conformal prediction

## 1 Introduction

Human motion forecasting—predicting future 3D pose trajectories from a short observed prefix—is a core primitive for human–robot interaction, animation, and anticipatory perception. Recent forecasters achieve strong point accuracy (e.g., mean per-joint position error (MPJPE)) with excellent runtime [18, 36], but point forecasts alone are insufficient for downstream decision-making. Reliable systems must reason about *uncertainty*: multiple futures may be plausible due to intrinsic ambiguity (aleatoric), and models may become unreliable when extrapolating

beyond their training support (epistemic), especially under distribution shift [23, 38].

Bayesian deep learning provides a principled lens for epistemic uncertainty, yet practical approaches often rely on approximate inference and repeated stochastic forward passes (e.g., Monte Carlo (MC) dropout [15] or ensembles [26]) and can degrade under shift [38]. Moreover, modern neural networks are frequently miscalibrated even in independent and identically distributed (i.i.d.) settings, motivating calibration layers with explicit guarantees [17]. Neural-linear / Bayesian last-layer methods offer a pragmatic alternative: keep a deterministic feature extractor and place a conjugate Bayesian model on the final linear layer, yielding closed-form posterior predictive uncertainty at near single-pass cost [12, 32, 39, 45, 49]. However, neural-linear uncertainty is typically paired with factorized (diagonal or isotropic) observation models [12, 24, 32], making it nontrivial to preserve structured spatio-temporal correlations for trajectory outputs.

In parallel, conformal prediction provides distribution-free predictive inference with finite-sample marginal validity under exchangeability [1, 28, 48]. Conformal methods have also been extended to functional/trajectory outputs, yielding finite-sample valid prediction bands (tubes) around a base forecaster [13, 29]. For regression, conformalization can wrap any base predictor (including heteroscedastic quantile regressors [40]) to guarantee target coverage, but it does not by itself separate epistemic from aleatoric uncertainty, and achieving strong conditional guarantees remains challenging in general [16].

We bridge two practical gaps with a lightweight *Bayesian-conformal hybrid* for motion forecasting: (i) adding an analytic epistemic signal while preserving structured spatio-temporal covariance for trajectory outputs, and (ii) providing a final distribution-free marginal calibration layer for prediction tubes. Starting from a strong deterministic MLP backbone, we place a conjugate Bayesian model on the final linear layer and derive an analytic, input-dependent *epistemic scale*  $\kappa(x)$  (or timewise  $\kappa_t(x)$ ) with a leverage-style interpretation. We use  $\kappa$  to multiplicatively inflate a structured Gaussian aleatoric covariance that models spatio-temporal and skeletal correlations, and we apply split conformal calibration to obtain 95% prediction tubes with finite-sample *marginal* coverage guarantees under exchangeability [13, 29], complementing Bayesian credible regions under model misspecification [14, 55]. Beyond global tube calibration, we also introduce a  $\kappa$ -conditioned (Mondrian) conformal variant for risk-adaptive tube scaling and a trajectory-level conformal ellipsoid using the structured covariance (Sec. 3.2).

*Contributions.* We contribute:

- **Closed-form epistemic scaling for motion forecasting.** A conjugate Bayesian last-layer head that produces an analytic epistemic factor  $\kappa(x)$  (or  $\kappa_t(x)$ ) that increases uncertainty for feature vectors far from the training design. In the multi-output conjugate model, conditioning on a positive-definite structured covariance  $\Sigma_{\text{str},t}(x)$ , the posterior predictive covariance factorizes exactly as  $\kappa_t(x) \Sigma_{\text{str},t}(x)$  (Prop. 1), justifying scalar inflation of structured covariances without Monte Carlo sampling.

- **Hybrid structured uncertainty.** A principled combination of epistemic scaling with a structured Gaussian aleatoric head that models spatio-temporal correlations (including a graph-coupled coordinate factor), yielding interpretable and efficient trajectory uncertainty.
- **Conformal calibration for tubes and sets.** Split conformal calibration of marginal prediction tubes to guarantee target coverage in finite samples, complementing Bayesian credible regions under misspecification [14, 55]. We further introduce (i)  $\kappa$ -conditioned (Mondrian) conformal tubes for risk-adaptive calibration and  $\kappa$ -binned coverage/width reporting [4, 47] and (ii) a trajectory-level conformal ellipsoid based on a structured Mahalanobis/whitened-residual score under  $\Sigma_T^{\text{hyb}} \otimes \Sigma_C$  (see supplementary material).
- **Broad empirical gains with deployment relevance.** Across nine datasets, *ConjGraph-CP* achieves the best average NLL rank and best overall MPJPE+NLL rank (mean of MPJPE/NLL ranks) while keeping uncertainty inference single-pass (no MC sampling loops), yielding a practical accuracy–reliability trade-off for real-time stacks.
- **Actionable epistemic risk monitoring.** We show that  $\kappa$  stratifies forecast error and supports selective deployment via abstention/fallback on high- $\kappa$  windows: on Human3.6M, the highest- $\kappa$  decile has  $1.79\times$  MPJPE compared to the average, and filtering that decile reduces MPJPE by  $\approx 9\%$  (see supplementary material; thresholds tuned on held-out data, with no formal guarantee implied).

## 2 Related Work

Human motion forecasting has long been dominated by deterministic point predictors, ranging from early recurrent/seq2seq models [36] to modern feed-forward backbones that offer strong runtime–accuracy trade-offs [18]. Recent deterministic improvements inject additional structure via graph convolutions on the kinematic tree [10], auxiliary supervision [51], stability-inducing integration schemes [6], or refinement via deviation feedback [44]. To represent the inherent ambiguity of future motion, multimodal predictors pursue latent-variable and generative formulations, including normalizing flows [53], diverse candidate generation [34], and diffusion-style denoising models [7]. While these approaches can capture multiple plausible futures, their inference often involves sampling or multiple hypotheses, which motivates lightweight uncertainty layers that preserve real-time point-forecasting backbones [7, 34, 53].

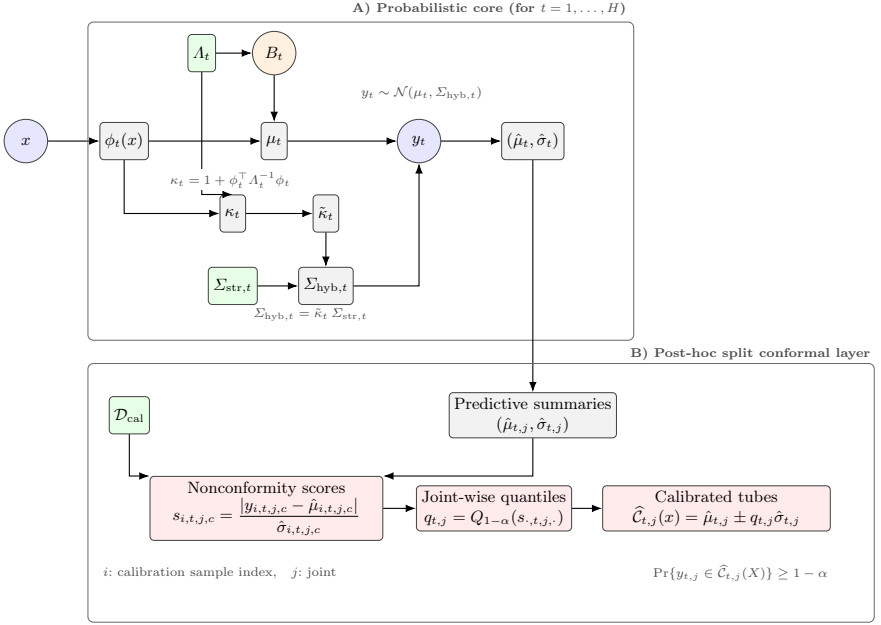
Uncertainty quantification for deep regression typically distinguishes aleatoric ambiguity from epistemic uncertainty due to limited training support [23]. Common epistemic estimators such as MC dropout and deep ensembles [15, 26] can improve robustness but require repeated stochastic forward passes and can still fail under distribution shift [38]. Neural-linear and last-layer Bayesian approximations provide a pragmatic alternative: keep the feature extractor deterministic and place Bayesian inference on the final linear layer, yielding analytic predictive uncertainty at near single-pass cost [12, 32, 39, 45, 49]. However, these epistemic

estimators are frequently used with factorized (diagonal or isotropic) likelihoods in regression [12, 24, 32], which does not capture spatio-temporal or skeletal correlations needed for coherent trajectory tubes and likelihood.

Structured Gaussian models such as matrix-normal/Kronecker factorizations provide a classical route to capturing separable spatio-temporal correlations, and graph/Gaussian Markov random field (GMRF) constructions encode skeletal coupling via sparse precisions [3, 11, 41]. Separately, conformal prediction offers distribution-free predictive inference with finite-sample *marginal* validity under exchangeability [1, 28, 48] and has been extended to prediction bands/tubes for functional and trajectory outputs [13, 29]; for regression, conformalized quantile regression is a strong calibration baseline [40], but conditional guarantees generally require additional assumptions [16, 47]. For dependent time series, sequential conformal inference updates intervals online using feedback and can target asymptotic conditional coverage under additional assumptions [50]. Conformal calibration has also been used to construct uncertainty sets for trajectory prediction under distribution shift and to provide probabilistic safety guarantees in planning via conformal prediction regions [21, 30]. In 3D human motion forecasting, latent conformal prediction has been proposed to calibrate multidimensional trajectory uncertainty [31]. More broadly, conformal layers have been paired with approximate Bayesian posteriors, including posterior-aware conformal intervals via last-layer Laplace [24] and posterior conformal prediction [55]. Taken together, these works suggest a practical design for motion forecasting pipelines that need uncertainty as a *drop-in* upgrade: we want (i) an analytic, input-dependent epistemic signal that can multiplicatively scale an arbitrary structured spatio-temporal covariance and (ii) a final distribution-free calibration layer for the resulting trajectory tubes. Building on these strands, we propose a Bayesian–conformal hybrid for motion forecasting that combines a conjugate Bayesian last-layer scale  $\kappa$  with a structured Gaussian aleatoric head and split conformal calibration, yielding interpretable epistemic inflation and calibrated 95% prediction tubes that work as a drop-in upgrade for strong deterministic forecasters [14, 55]. Moreover, we show how the analytic  $\kappa$  signal can be used *inside* the conformal layer via  $\kappa$ -conditioned (Mondrian) calibration and how the structured covariance enables a trajectory-level conformal ellipsoid as an alternative to marginal tubes.

### 3 Method

We describe *ConjGraph-CP*, our default Bayesian–conformal hybrid that produces calibrated marginal prediction tubes. Figure 1 summarizes the dependency structure. Algorithm 1 gives an end-to-end recipe as a drop-in uncertainty upgrade for deterministic forecasters. We additionally describe two conformal variants enabled by the analytic  $\kappa$  and structured covariance:  $\kappa$ -conditioned (Mondrian) tubes and a trajectory-level conformal ellipsoid (Sec. 3.2). Simpler variants in our ablation are recovered by dropping the graph coupling, the timewise shrinkage, or the hybrid inflation.



**Fig. 1:** Formal dependency view of ConjGraph-CP. (A) Probabilistic core (plate over  $t$ ): deterministic features  $\phi_t(x)$ , conjugate Bayesian last-layer scale  $\kappa_t = 1 + \phi_t^\top \Lambda_t^{-1} \phi_t$ , hybrid covariance  $\Sigma_{\text{hyb},t} = \tilde{\kappa}_t \Sigma_{\text{str},t}$ , and predictive model  $y_t \sim \mathcal{N}(\mu_t, \Sigma_{\text{hyb},t})$ . (B) Post-hoc split conformal layer: calibration scores on  $\mathcal{D}_{\text{cal}}$  produce joint-wise quantiles  $q_{t,j}$  and calibrated tubes  $\hat{\mathcal{C}}_{t,j}(x) = \hat{\mu}_{t,j} \pm q_{t,j} \hat{\sigma}_{t,j}$  with finite-sample marginal validity.

### 3.1 Hybrid structured Gaussian predictor (ConjGraph)

Let  $x_{1:T} \in \mathbb{R}^{T \times C}$  be an observed pose prefix and  $y_{1:H} \in \mathbb{R}^{H \times C}$  the future trajectory, with  $C = 3J$  coordinates (flattened joints), where  $J$  is the number of joints. As in our implementation, we train and evaluate on *displacements* relative to the last observed pose  $x_T$  (i.e.,  $y_t \leftarrow y_t - x_T$ ); absolute poses are recovered by adding  $x_T$  back at inference.

*Goal: calibrated marginal tubes.* Our target object is a prediction tube  $\mathcal{C}_\alpha(x)$  with miscoverage  $\alpha \in (0, 1)$ . We primarily focus on axis-aligned marginal tubes of the form

$$\hat{\mathcal{C}}_\alpha(x) = \prod_{t=1}^H \prod_{j=1}^J \prod_{d=1}^3 [\hat{\mu}_{t,j,d}(x) \pm q_{t,j} \hat{\sigma}_{t,j,d}^{\text{CP}}(x)], \quad (1)$$

where  $(\hat{\mu}, \hat{\sigma}^{\text{CP}})$  come from a probabilistic predictor and  $\{q_{t,j}\}$  are split-conformal calibration factors (Sec. 3.2). We additionally consider a trajectory-level conformal set based on a structured Mahalanobis/whitened-residual score (Sec. 3.2).

*Backbone and time-domain design vector.* We follow the siMLPe/DCT-bridge backbone [18]: the input sequence is transformed along the time axis into a

discrete cosine transform (DCT) coefficient representation and passed through a deterministic trunk  $f_\theta$  that outputs coefficient-wise features  $h_n(x) \in \mathbb{R}^C$  for  $n \in \{1, \dots, N\}$ . Let  $A \in \mathbb{R}^{N \times N}$  denote the inverse DCT (iDCT) matrix that maps coefficient outputs back to the time domain. In the fully-connected output (FC-Out) parameterization used throughout this work, the time-domain mean at horizon step  $t$  is linear in an augmented design vector  $\phi_t(x) \in \mathbb{R}^P$ :

$$\hat{\mu}_t(x) = B_t^\top \phi_t(x), \quad (2)$$

$$\phi_t(x) = \begin{bmatrix} g_t(x) \\ s_t \end{bmatrix} \in \mathbb{R}^P, \quad P = C + 1,$$

$$g_t(x) = \sum_{n=1}^N A_{t,n} h_n(x) \in \mathbb{R}^C, \quad s_t = \sum_{n=1}^N A_{t,n} \in \mathbb{R}. \quad (3)$$

The scalar  $s_t$  appears because a coefficient-space bias becomes time-dependent after iDCT; in our implementation  $g_{1:H}(x)$  is computed by multiplying trunk coefficient features with the first  $H$  rows of  $A$ , and  $s_{1:H}$  is the corresponding iDCT row-sum used to propagate the bias term.

*Conjugate Bayesian last-layer scale.* We place a conjugate Bayesian model on the final linear map in (2) while keeping the trunk deterministic. For each horizon step  $t$ , consider the multi-output linear-Gaussian model

$$y_t = B_t^\top \phi_t(x) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma_{\text{str},t}(x)), \quad (4)$$

with a matrix-normal prior [11]

$$B_t \mid \Sigma_{\text{str},t} \sim \mathcal{MN}(B_{0,t}, \Lambda_0^{-1}, \Sigma_{\text{str},t}), \quad \Lambda_0 = \lambda_0 I_P. \quad (5)$$

**Proposition 1 (Exact structured-covariance scaling).** *Conditioning on any positive-definite  $\Sigma_{\text{str},t}(x)$ , integrating out  $B_t$  under the conjugate matrix-normal model in (4)–(5) preserves the structured covariance up to a scalar multiplicative inflation:*

$$y_t \mid x, \mathcal{D} \sim \mathcal{N}(\mu_t^{\text{post}}(x), \kappa_t(x) \Sigma_{\text{str},t}(x)), \quad \kappa_t(x) = 1 + \phi_t(x)^\top \Lambda_{n,t}^{-1} \phi_t(x),$$

where  $\Lambda_{n,t} = \Lambda_0 + \sum_{i \in \mathcal{D}} \phi_t(x_i) \phi_t(x_i)^\top$ . In particular,  $\kappa_t$  rescales only the covariance magnitude and preserves the correlation/structure encoded by  $\Sigma_{\text{str},t}$ .

*Proof.* By conjugacy,  $B_t \mid \Sigma_{\text{str},t}, \mathcal{D} \sim \mathcal{MN}(B_{n,t}, \Lambda_{n,t}^{-1}, \Sigma_{\text{str},t})$ , hence  $\text{vec}(B_t) \mid \Sigma_{\text{str},t}, \mathcal{D}$  is Gaussian with covariance  $\Sigma_{\text{str},t} \otimes \Lambda_{n,t}^{-1}$ . Therefore  $B_t^\top \phi_t(x)$  has covariance  $(\phi_t(x)^\top \Lambda_{n,t}^{-1} \phi_t(x)) \Sigma_{\text{str},t}(x)$ , and adding  $\varepsilon_t \sim \mathcal{N}(0, \Sigma_{\text{str},t}(x))$  yields the stated inflation.

*Remark 1 (Arbitrary structured covariance).* The factorization in Prop. 1 holds for any positive-definite  $\Sigma_{\text{str},t}(x)$  (dense, Kronecker-factored, or defined via a sparse precision), enabling epistemic inflation without simplifying the structured aleatoric head.

We use  $\kappa_t(x)$  as an analytic epistemic inflation factor (no MC sampling) that increases when the test feature vector  $\phi_t(x)$  has large leverage under the training design. In ConjGraph-CP we keep the backbone mean predictor  $\hat{\mu}_t(x)$  and use only  $\kappa_t(x)$  (and its stabilized version  $\tilde{\kappa}_t(x)$  below) to rescale the structured covariance head.

*Closed-form fitting and stabilization.* We only require  $\Lambda_{n,t}^{-1}$  to compute  $\kappa_t$ . With  $\Lambda_0 = \lambda_0 I_P$ , we fit  $\Lambda_{n,t}^{-1}$  in closed form by accumulating the sufficient statistic  $S_{xx,t} = \sum_i \phi_t(x_i) \phi_t(x_i)^\top$  on the training set and computing  $(S_{xx,t} + \lambda_0 I_P)^{-1}$  once via a Cholesky factorization. At inference, we compute  $\kappa_t(x) = 1 + \phi_t(x)^\top \Lambda_{n,t}^{-1} \phi_t(x)$ ; equivalently,  $\kappa_t(x) = 1 + \|L_{n,t}^{-1} \phi_t(x)\|_2^2$  using the Cholesky factor  $L_{n,t} L_{n,t}^\top = \Lambda_{n,t}$ . In our implementation we store  $\Lambda_{n,t}^{-1}$  (via a Cholesky inverse) and evaluate the quadratic form directly, avoiding MC sampling. To stabilize horizon-wise scaling, we optionally shrink  $\kappa_t$  toward a time-homogeneous sequence-level scale derived from pooled statistics. Let  $\Lambda_{\text{glob}} = \lambda_0 I_P + \sum_{t,i} \phi_t(x_i) \phi_t(x_i)^\top$  and define

$$\bar{\kappa}(x) = \frac{1}{H} \sum_{t=1}^H \left( 1 + \phi_t(x)^\top \Lambda_{\text{glob}}^{-1} \phi_t(x) \right). \quad (6)$$

Our stabilized scale is

$$\tilde{\kappa}_t(x) = (1 - \rho) \kappa_t(x) + \rho \bar{\kappa}(x), \quad \rho \in [0, 1]. \quad (7)$$

The global- $\kappa$  variant is recovered by setting  $\rho = 1$ . In experiments we also allow a monotone “epistemic temperature” transform  $\tilde{\kappa}_t \leftarrow \tilde{\kappa}_t^\gamma$  with  $\gamma > 0$  (with  $\gamma = 1$  corresponding to the default construction) to tune the NLL/interval-width (mean marginal interval width; reported as W95) trade-off.

*Joint-wise leverage scale for CP tubes.* The global timewise scale  $\tilde{\kappa}_t(x)$  is shared across all joints at horizon  $t$ . To allow joint-dependent epistemic inflation in the CP layer without changing the likelihood, we compute an additional per-joint scale  $\kappa_j(x)$  from reduced per-joint design vectors. Let  $\phi_{t,j}(x) \in \mathbb{R}^4$  contain the 3 coordinates of joint  $j$  in  $g_t(x)$  plus the shared bias feature  $s_t$ . We fit a joint-wise conjugate statistic  $\Lambda_{n,j}^{-1} = (\lambda_{0,\text{joint}} I + \sum_{i,t} \phi_{t,j}(x_i) \phi_{t,j}(x_i)^\top)^{-1}$  on  $\mathcal{D}_{\text{train}}$  and define  $\kappa_{t,j}(x) = 1 + \phi_{t,j}(x)^\top \Lambda_{n,j}^{-1} \phi_{t,j}(x)$ , aggregated as  $\kappa_j(x) = \frac{1}{H} \sum_{t=1}^H \kappa_{t,j}(x)$ . For conformal tubes we use

$$\hat{\sigma}_{t,j,d}^{\text{CP}}(x) = \sqrt{\kappa_j(x)} \hat{\sigma}_{t,j,d}(x), \quad (8)$$

while the probabilistic core (and NLL) remains unchanged.

*Structured Gaussian aleatoric covariance (MN-GraphJ).* We refer to this matrix-normal + joint-graph construction as *MN-GraphJ* (MatrixNormal-GraphJ). The backbone predicts a structured *aleatoric* covariance for the residual matrix  $R(x) = y_{1:H} - \hat{\mu}_{1:H}(x) \in \mathbb{R}^{H \times C}$ . We use a separable matrix-normal parameterization,

$$\text{vec}(R(x)) \sim \mathcal{N}(0, \Sigma_T(x) \otimes \Sigma_C(x)), \quad \text{equivalently} \quad R(x) \sim \mathcal{MN}(0, \Sigma_T(x), \Sigma_C(x)), \quad (9)$$



where  $\Sigma_T(x) \in \mathbb{R}^{H \times H}$  captures temporal correlations and  $\Sigma_C(x) \in \mathbb{R}^{C \times C}$  captures spatial/skeletal coupling. We parameterize  $\Sigma_T(x) = L_T(x)L_T(x)^\top$  with a learned lower-triangular  $L_T(x)$  (packed Cholesky representation), enabling dense temporal correlations with stable optimization. To encode skeletal structure, we model  $\Sigma_C$  via a joint-graph precision (a GMRF construction [3, 41]). Let  $L_{\text{joint}} \in \mathbb{R}^{J \times J}$  be the unweighted graph Laplacian of a chosen joint graph (e.g., the kinematic tree, or a data-driven  $k$ -nearest-neighbor (kNN) graph estimated from training motions). The network predicts positive scalars  $\tau(x)$  and  $\epsilon(x)$ , and we define

$$Q_J(x) = \tau(x) L_{\text{joint}} + \epsilon(x) I_J, \quad \Sigma_C(x) = (Q_J(x) \otimes I_3)^{-1}. \quad (10)$$

*Hybridization.* We combine epistemic and aleatoric components by inflating the temporal covariance factor with  $\tilde{\kappa}_t(x)$ . Define  $D_\kappa(x) = \text{diag}(\sqrt{\tilde{\kappa}_1(x)}, \dots, \sqrt{\tilde{\kappa}_H(x)})$  and set

$$\Sigma_T^{\text{hyb}}(x) = D_\kappa(x) \Sigma_T(x) D_\kappa(x)^\top, \quad \Sigma^{\text{hyb}}(x) = \Sigma_T^{\text{hyb}}(x) \otimes \Sigma_C(x). \quad (11)$$

Since  $D_\kappa$  is diagonal,  $D_\kappa^\top = D_\kappa$ ; we write the transpose to make the intended similarity transform explicit. Equivalently, if  $\Sigma_T(x) = L_T(x)L_T(x)^\top$ , we implement (11) by scaling the  $t$ -th row of  $L_T(x)$  by  $\sqrt{\tilde{\kappa}_t(x)}$  (row-wise scaling), which preserves correlation structure while inflating scale. The resulting marginal variance factorizes as

$$\hat{\sigma}_{t,c}^2(x) = \text{Var}(y_{t,c} \mid x) = \tilde{\kappa}_t(x) [\Sigma_T(x)]_{tt} [\Sigma_C(x)]_{cc}. \quad (12)$$

### 3.2 Split conformal calibration for prediction tubes

Bayesian credible regions can be miscalibrated under model misspecification or dataset shift [14, 55]. We therefore conformalize the hybrid Gaussian predictor to obtain calibrated marginal tubes with finite-sample validity under exchangeability [1, 28, 48].

On a held-out calibration set  $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$ , we compute standardized absolute residual scores

$$s_{i,t,j,d} = \frac{|y_{i,t,j,d} - \hat{\mu}_{t,j,d}(x_i)|}{\hat{\sigma}_{t,j,d}^{\text{CP}}(x_i)}, \quad (13)$$

where  $\hat{\sigma}^{\text{CP}}$  is the scale used for tube construction (in ConjGraph-CP this includes the hybrid timewise inflation  $\tilde{\kappa}_t$  and the CP-only joint factor in (8)). For each horizon step  $t$  and joint  $j$ , we set  $q_{t,j}$  using the split-conformal “higher” quantile rule,

$$q_{t,j} = \text{Quantile}(\{s_{i,t,j,d}\}_{i,d}, \lceil (N_{t,j} + 1)(1 - \alpha) \rceil / N_{t,j}), \quad (14)$$

where  $N_{t,j}$  is the number of pooled scores for  $(t, j)$  (in our implementation  $N_{t,j} = n \cdot 3$ ). We then output the tube in (1).



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**Algorithm 1** ConjGraph-CP: single-pass structured uncertainty with analytic  $\kappa$  and split conformal calibration

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- Require:** Training data  $\mathcal{D}_{\text{train}}$ , calibration data  $\mathcal{D}_{\text{cal}}$ , miscoverage  $\alpha$ , prior precisions  $\lambda_0$  and  $\lambda_{0,\text{joint}}$
- 1: Train a deterministic mean forecaster (mean fine-tune; MeanFT) to obtain time-domain features  $\phi_t(x)$  and mean predictions  $\hat{\mu}_t(x)$ .
  - 2: Fit a structured Gaussian covariance head (e.g., MN-GraphJ) on residuals with frozen mean, yielding  $\Sigma_{\text{str},t}(x)$ .
  - 3: Accumulate  $S_{xx,t} \leftarrow \sum_{(x_i, \cdot) \in \mathcal{D}_{\text{train}}} \phi_t(x_i) \phi_t(x_i)^\top$  and compute  $\Lambda_{n,t}^{-1} = (\lambda_0 I + S_{xx,t})^{-1}$  (e.g., via Cholesky factorization).
  - 4: Fit joint-wise leverage stats: accumulate  $S_{xx,j} \leftarrow \sum_{(x_i, \cdot) \in \mathcal{D}_{\text{train}}, t} \phi_{t,j}(x_i) \phi_{t,j}(x_i)^\top$  and compute  $\Lambda_{n,j}^{-1} = (\lambda_{0,\text{joint}} I + S_{xx,j})^{-1}$ . Use it to compute  $\kappa_j(x)$  and  $\hat{\sigma}^{\text{CP}}$  via (8).
  - 5: **Split conformal calibration:** on  $\mathcal{D}_{\text{cal}}$ , compute  $s_{i,t,j,d} = |y_{i,t,j,d} - \hat{\mu}_{t,j,d}(x_i)| / \hat{\sigma}_{t,j,d}^{\text{CP}}(x_i)$  and set  $q_{t,j} \leftarrow \text{Quantile}(\{s_{i,t,j,d}\}_{i,d}, \lceil (N_{t,j} + 1)(1 - \alpha) \rceil / N_{t,j})$  (higher), where  $N_{t,j}$  is the number of pooled scores for  $(t, j)$ .
  - 6: **(Optional)  $\kappa$ -Mondrian CP:** bin sequences by  $r(x)$  (e.g., mean  $\sqrt{\kappa_t(x)}$ ) and compute  $q_{t,j,b}$  per bin  $b$ ; at test time use  $q_{t,j,b(x)}$ .
  - 7: **(Optional) trajectory CP:** compute  $s_i = \|L_T^{\text{hyb}}(x_i)^{-1}(y_i - \hat{\mu}(x_i))L_C(x_i)^{-\top}\|_F$  and conformalize to get  $q_{\text{traj}}$ , yielding an ellipsoidal set  $\{y : s(y) \leq q_{\text{traj}}\}$ .
  - 8: **Prediction for new  $x$ :** compute  $\kappa_t(x) = 1 + \phi_t(x)^\top \Lambda_{n,t}^{-1} \phi_t(x)$ , optionally shrink to  $\tilde{\kappa}_t(x)$  via (7), and form  $\Sigma_{\text{hyb},t}(x) = \tilde{\kappa}_t(x) \Sigma_{\text{str},t}(x)$ .
  - 9: Output calibrated tube  $\hat{\mathcal{C}}_\alpha(x)$  in (1) (or  $q_{t,j,b(x)}$ ), and optionally the trajectory ellipsoid.
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*Guarantee and practical notes.* Split conformal provides finite-sample *marginal* validity for the induced score distribution under exchangeability. Pooling over the 3 coordinates per joint yields stable joint-wise scales  $q_{t,j}$  and targets the mean marginal coverage metric we report; per-coordinate  $q_{t,j,d}$  or max-score calibration yields stricter (but more conservative) validity notions.

*Risk-adaptive and set-valued variants (supplementary).* We also consider (i)  $\kappa$ -conditioned (Mondrian) conformal tubes that bin sequences by a summary of  $\tilde{\kappa}_t(x)$  and calibrate bin-wise  $q_{t,j,b}$  [4, 47], and (ii) a trajectory-level conformal ellipsoid based on a structured Mahalanobis/whitened-residual score under  $\Sigma_T^{\text{hyb}} \otimes \Sigma_C$ ; details and additional metrics are in the supplementary material.

## 4 Experiments

### 4.1 Experimental setup

*Datasets.* We treat Human3.6M (H36M) [22] as the primary benchmark (indoor motion capture (MoCap)) and additionally evaluate cross-dataset generalization on AMASS [33], LaFAN1 [19], CMU-MoCap [5], 3DPW [35], and three human-robot interaction (HRI)-style datasets with different capture / interaction setups: CHICO [42], HA4M [8], and AnDy [37]. Unless explicitly stated otherwise, non-H36M datasets use  $T=50$  observed frames and we forecast  $H=25$  future frames.

*Evaluation protocol.* We use a random-but-deterministic split of the *test* windows: we fix the random number generator (RNG) seed to 304, permute the test windows, use the first  $n_{\text{cal}}=512$  windows for split conformal calibration, and report all metrics on the next  $n_{\text{eval}}=1024$  windows. We use  $\alpha=0.05$  throughout. Because the evaluation indices depend on  $n_{\text{cal}}$ , we do not change  $(n_{\text{cal}}, n_{\text{eval}}, \alpha)$  across models.

*Metrics.* We report mean per-joint position error (MPJPE), final displacement error (FDE; here implemented as final-horizon MPJPE), and per-element Gaussian negative log-likelihood (NLL) (averaged over horizons and coordinates). For space, we present MPJPE/FDE as “MPJPE / FDE” in the main tables. Because absolute metric scales and units vary across datasets, we additionally report per-dataset ranks in cross-dataset summaries; MPJPE+NLL rank denotes the mean of MPJPE and NLL ranks (Table 1). For uncertainty, we evaluate nominal two-sided 95% marginal intervals  $\hat{\mu} \pm Z_{0.95} \hat{\sigma}$  (**Cov95/W95**) and split-conformalized (conformal prediction; CP) intervals  $\hat{\mu} \pm q_{t,j} \hat{\sigma}^{\text{CP}}$  (**Cov95(CP)/W95(CP)**), where  $Z_{0.95} = \Phi^{-1}(0.975) \approx 1.960$  and  $\Phi$  is the standard normal cumulative distribution function. Cov95 is empirical marginal coverage (fraction of scalar targets  $(t, c)$  covered), and W95 is the corresponding mean marginal interval width (averaged over  $(t, c)$ ). Conformal scales  $\{q_{t,j}\}$  are calibrated *per horizon and joint* by pooling standardized residuals  $|y_{t,j,d} - \hat{\mu}_{t,j,d}| / \hat{\sigma}_{t,j,d}^{\text{CP}}$  across the 3 coordinates  $d$  of each joint on the calibration split. Deterministic models are paired with a fixed isotropic observation noise  $\sigma_{\text{obs}}=0.017\text{m}$  to compute NLL and nominal intervals; they do not output  $\hat{\sigma}_{t,c}(x)$  and thus do not report conformal tube metrics. To evaluate risk-adaptive and set-valued variants beyond marginal tubes, we additionally report (i)  $\kappa$ -binned coverage/width for  $\kappa$ -conditioned (Mondrian) conformal tubes and (ii) a trajectory-level conformal ellipsoid based on the structured Mahalanobis/whitened-residual score, including a log-volume-per-dimension efficiency metric (details in supplementary material).

*Training and calibration.* All methods share the siMLPe/DCT backbone [18] (MLP in DCT coefficient space, mapped to the time domain via iDCT; we forecast displacements relative to  $x_T$  and add  $x_T$  back at inference, as described in Sec. 3.1). All methods are trained with Adam; for the siMLPe backbone we use 2000 steps for base training and an 800-step mean fine-tune (MeanFT), and probabilistic heads are trained for 800 steps with frozen mean (batch sizes 256/128/64, respectively). For structured Gaussian heads, we initialize the mean from a trained MeanFT model and (when applicable) freeze the mean while optimizing only covariance parameters via NLL. For **ConjGraph-CP**, we fit the conjugate-Bayes last-layer statistics  $\Lambda_{n,t}^{-1}$  in closed form on the training split (no gradient steps), compute timewise  $\kappa_t(x) = 1 + \phi_t(x)^\top \Lambda_{n,t}^{-1} \phi_t(x)$ , and use the shrinkage/temperature stabilizers  $(\rho, \gamma; \text{Sec. 3.1})$  as lightweight knobs to control horizon-wise variability and the NLL–W95 trade-off. We select  $(\lambda_0, \rho, \gamma)$  per dataset on held-out training windows by minimizing a penalized NLL objective with soft constraints on MPJPE and calibration (coverage/width), and then

keep them fixed for all reported evaluations. Finally, we compute  $\{q_{t,j}\}$  on the calibration split and evaluate calibrated tubes on the evaluation split.

## 4.2 Baselines

We compare against baseline families evaluated under the same protocol: (i) deterministic point forecasters (siMLPe/MeanFT and recent variants such as AuxTasks [51], Symplectic [6], DeFeeNet [44], and HumanMAC [7], plus graph-based SeSGCN [42]); (ii) sampling/ensemble uncertainty baselines (deep ensembles [26] and multi-head predictors such as GSPS [34]); (iii) calibration baselines (Bridge-CQR: conformalized quantile regression [40]); and (iv) Gaussian uncertainty heads (diagonal DCT-Bridge; separable matrix-normal [11]; and graph/GMRF-style structured covariances [3, 41]); and (v) external stochastic/diffusion forecasters (SkeletonDiffusion [9], BeLFusion [2], TransFusion [46], CoMusion [43], SPARD [54], MotionMap [20], and SLD-HMP [52]). When external methods provide public checkpoints, we initialize from the released weights and fine-tune under our protocol; if checkpoints exist only for a subset of datasets, we initialize from the closest available one (e.g., Human3.6M/AMASS) and fine-tune, otherwise we train from scratch.

## 4.3 Cross-dataset comparison

We report a broader cross-dataset comparison to contextualize performance beyond Human3.6M. Bold (underline) indicates best (second-best) among the shown methods (ties included); for coverage, best/second-best is defined by closeness to the 0.95 target. Because absolute metric scales and units vary strongly across datasets, we summarize performance with mean ranks, denoted MR (Table 1;  $\downarrow$  better); full per-dataset metrics are deferred to the supplementary material (Sec. *Cross-dataset detailed results*). Overall rank (Table 1) shows ConjGraph-CP achieves the best  $\text{MR}_{\text{MPJPE}+\text{NLL}}$  while also giving the best  $\text{MR}_{\text{NLL}}$ , with the additional interpretability of  $\kappa$  discussed below.

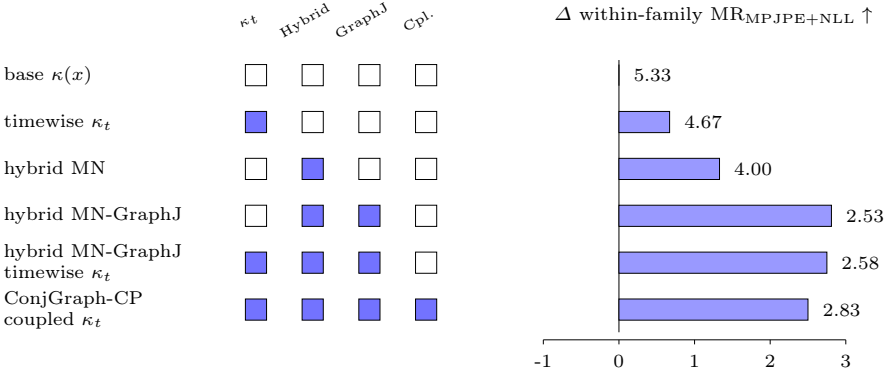
*Practical benefits over existing methods.* ConjGraph-CP provides four concrete gains in this benchmark. First, it gives the strongest *accuracy-likelihood balance*:  $\text{MR}_{\text{MPJPE}+\text{NLL}} = 2.72$  with  $\text{MR}_{\text{MPJPE}} = 4.44$  and  $\text{MR}_{\text{NLL}} = 1.00$  (Table 1). Second, it delivers strong probabilistic fit: best NLL among shown methods on every dataset block while retaining competitive MPJPE (supplementary, Sec. *Cross-dataset detailed results*). Third, its calibrated tube efficiency is strong (best  $\text{MR}_{W95(CP)} = 2.28$ ), so uncertainty quality does not come from indiscriminately widening intervals. Fourth,  $\kappa$  provides an actionable single-pass risk signal beyond average metrics: it stratifies error monotonically and enables selective deployment via abstention/fallback on high- $\kappa$  windows (see supplementary material; thresholds tuned on held-out data, with no formal guarantee implied).

To isolate the contributions of structured covariance, epistemic scaling, and time-coupled  $\kappa_t$  shrinkage, we summarize an FC-Out component ablation across datasets in Fig. 2; full per-dataset results are deferred to the supplementary material.

**Table 1:** Mean rank (MR;  $\downarrow$  better) across datasets. Ranks are computed per dataset and averaged over available datasets.  $\text{MR}_{\text{MPJPE}+\text{NLL}}$  is the mean of MPJPE and NLL ranks (available for all models).

| Model                             | $\text{MR}_{\text{MPJPE}} \downarrow$ | $\text{MR}_{\text{NLL}} \downarrow$ | $\text{MR}_{W95(\text{CP})} \downarrow$ | $\text{MR}_{\text{MPJPE}+\text{NLL}} \downarrow$ |
|-----------------------------------|---------------------------------------|-------------------------------------|-----------------------------------------|--------------------------------------------------|
| siMLPe (base)                     | 7.89                                  | 10.67                               | —                                       | 9.28                                             |
| MeanFT (siMLPe fine-tune)         | <b>2.22</b>                           | 6.11                                | —                                       | 4.17                                             |
| AuxTasks (50/50)                  | 9.06                                  | 11.78                               | —                                       | 10.42                                            |
| Symplectic (50/50)                | 16.22                                 | 15.78                               | —                                       | 16.00                                            |
| DeFeeNet (50/50)                  | 7.61                                  | 10.44                               | —                                       | 9.03                                             |
| GPS (50/50)                       | 7.67                                  | 9.00                                | 5.44                                    | 8.33                                             |
| HumanMAC (50/50)                  | 10.22                                 | 10.00                               | —                                       | 10.11                                            |
| SeSGCN (teacher)                  | 13.39                                 | 11.78                               | —                                       | 12.58                                            |
| Deep ensemble                     | 8.22                                  | 5.56                                | 5.39                                    | 6.89                                             |
| Bridge-CQR                        | 4.56                                  | <u>3.00</u>                         | 9.11                                    | <u>3.78</u>                                      |
| SkeletonDiffusion (CVPR 2025) [9] | 15.00                                 | 13.67                               | 9.78                                    | 14.33                                            |
| BeLFusion (ICCV 2023) [2]         | 9.56                                  | 8.00                                | 5.89                                    | 8.78                                             |
| TransFusion (RA-L 2024) [46]      | 12.44                                 | 9.78                                | 4.89                                    | 11.11                                            |
| CoMusion (ECCV 2024) [43]         | 6.06                                  | 6.44                                | 3.83                                    | 6.25                                             |
| SPARD (AAAI 2026) [54]            | 7.11                                  | 6.56                                | <u>3.44</u>                             | 6.83                                             |
| MotionMap (CVPR 2025) [20]        | 16.11                                 | 16.22                               | 9.44                                    | 16.17                                            |
| SLD-HMP (ECCV 2024) [52]          | 13.22                                 | 15.22                               | 6.50                                    | 14.22                                            |
| ConjGraph-CP <sup>†</sup>         | <u>4.44</u>                           | <b>1.00</b>                         | <b>2.28</b>                             | <b>2.72</b>                                      |

**Legend:** **bold** = best; underline = second-best (ties included). <sup>†</sup> (MN-GraphJ, time-coupled  $\kappa_t$ , shrink).



**Fig. 2:** FC-Out component ablation summary across datasets. Left: active components per variant. Right: improvement in within-family  $\text{MR}_{\text{MPJPE}+\text{NLL}}$  (computed only among these ablation variants; not comparable to Table 1) relative to the base  $\kappa(x)$  variant (higher is better); numbers annotate the absolute within-family  $\text{MR}_{\text{MPJPE}+\text{NLL}}$ .

#### 4.4 FC-Out ablation: component contributions

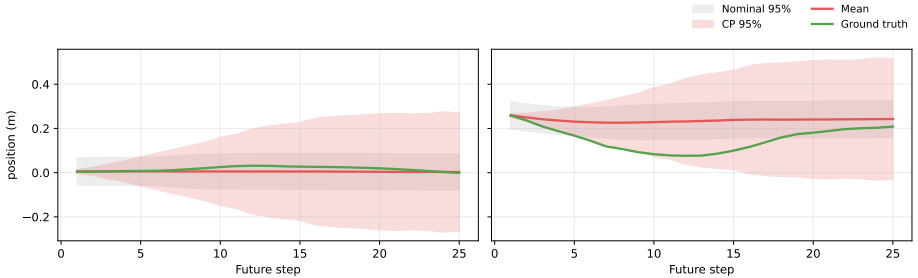
*Ablation findings (all datasets; full 9/9 coverage).* The best variant in this family is *hybrid MN-GraphJ* with  $\text{MR}_{\text{MPJPE}+\text{NLL}} = 2.53$ , followed by uncoupled timewise  $\kappa_t$  (2.58) and *ConjGraph-CP* (time-coupled  $\kappa_t$ ; 2.83) (Fig. 2). We still use *ConjGraph-CP* (time-coupled  $\kappa_t$ ) as the default because it is optimized for calibrated tube efficiency, achieving the best  $\text{MR}_{W95(\text{CP})}$  and best global  $\text{MR}_{\text{MPJPE}+\text{NLL}}$  on the full benchmark (Table 1). The main gains come from

hybrid structured covariance and GraphJ coupling: relative to basic FC-Out  $\kappa(x)$  (within-family  $\text{MR}_{\text{MPJPE}+\text{NLL}} = 5.33$ ), hybrid MN improves to 4.00, and GraphJ improves further to 2.53. Time-coupled shrinkage is a key knob for the likelihood–interval trade-off: within the family,  $\text{MR}_{\text{NLL}}$  improves from 2.56 (hybrid MN-GraphJ) to 2.44 (time-coupled  $\kappa_t$ ), and in the full benchmark it yields our strongest calibrated tube efficiency (Table 1). While the DCT-pinvar variant (not shown) is consistently unfavorable in this study (worst overall rank 6.11), it is not part of the recommended recipe.

*Compute.* All structured heads (including ConjGraph-CP) are single-pass at inference; ConjGraph-CP adds only horizon-wise quadratic forms for  $\kappa_t$  and does not require MC sampling. The closed-form fit of  $\Lambda_{n,t}^{-1}$  is an offline pass over training features, and split conformal calibration uses only 512 held-out test windows. Compute/memory overhead and latency measurements are reported in the supplementary material.

*Controlled synthetic benchmark (supplementary).* To validate component recovery under a known ground-truth uncertainty decomposition, we report a controlled synthetic benchmark in the supplementary material.

## 4.5 Qualitative uncertainty visualization



**Fig. 3:** Interpretable calibrated confidence intervals. We select two evaluation sequences with low vs. high mean epistemic scale  $\bar{\kappa}$  and plot a representative coordinate time series (chosen by maximum predicted uncertainty at the final horizon step). The high- $\bar{\kappa}$  case yields wider intervals. Gray: nominal 95% intervals ( $\pm 1.960\sigma$ ). Red: split-conformal calibrated marginal intervals ( $\pm q_{t,j}\sigma$ ), which restore the target 95% marginal coverage.

*Epistemic scale diagnostics (supplementary).* Additional diagnostics on how  $\kappa_t$  inflates uncertainty and how  $\kappa$  stratifies error are deferred to the supplementary material.

## 5 Discussion and limitations

In real-time risk-sensitive systems (e.g., human–robot interaction (HRI), human-aware navigation, or safety filters for embodied agents), planners often require predictive envelopes to enforce collision avoidance and personal-space

constraints [25, 27, 42]. Overconfident forecasters can therefore be dangerous: under novel motions or distribution shift, a point-accurate mean may come with an uncertainty tube that is too narrow, leading to unsafe commitments [23, 38]. ConjGraph-CP is designed for this setting:  $\kappa_t(x)$  provides a single-pass epistemic risk score that inflates a physically coherent spatio-temporal covariance, and split conformal calibration restores target marginal coverage in finite samples. Practically,  $\kappa$  can be used as a lightweight risk monitor (e.g., thresholding a sequence-level summary to trigger conservative fallbacks, with thresholds tuned on held-out calibration/validation data).

*Limitations.* The approach relies on the learned feature representation: if the feature extractor is poorly aligned with uncertainty-relevant novelty,  $\kappa$  may be less informative. Moreover, conformal coverage guarantees are marginal, and obtaining strong conditional guarantees remains an open challenge without additional assumptions [16]. Finally, the graph-structured covariance is a modeling choice; richer dependencies may require denser coordinate models at additional computational cost.

*Future directions.* Potential extensions include online updating of sufficient statistics for  $\kappa$  under streaming data and moving beyond marginal tubes when coherent set-valued prediction is required.

## 6 Conclusion

We proposed a hybrid uncertainty model for motion forecasting that combines a conjugate Bayesian last-layer epistemic scale with a structured Gaussian aleatoric covariance head and conformal calibration. The method provides interpretable epistemic scaling, efficient inference, and calibrated 95% prediction tubes with strong likelihood across diverse motion forecasting datasets. Based on the full ablation, our recommended default recipe is *ConjGraph-CP* (hybrid MN-GraphJ with time-coupled  $\kappa_t$  shrink), while simpler FC-Out and DCT-pinv variants are mainly useful as analysis baselines.

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